Hamiltonian and Hamilton-connected thresholds for planar recolouring graphs

Abstract

For a graph G, let $\mathcal{R}_k(G)$ have the proper colourings of G with a labelled k-colour palette as vertices, with one-vertex recolourings as edges. We prove that $\mathcal{R}_k(G)$ is Hamilton-connected and vertex-pancyclic whenever G is nonempty and d-degenerate and $k \geq d+3$; this uniform palette bound is sharp for both properties. We determine the common Hamiltonian and Hamilton-connected thresholds as 7 for planar graphs, 5 for triangle-free and bipartite planar graphs, and 4 for planar graphs of girth at least six and for forests. At four colours we also prove Hamiltonicity for planar graphs of girth at least five with at most seven distinct 5-cycles, and for those whose blocks all have maximum degree at most three. The latter condition allows unbounded degree at cutvertices. These four-colour proofs combine structural reductions with local restoration certificates supplied in a separate, reproducible supplement. For unrestricted planar graphs of girth at least five, both thresholds remain between 4 and 5.

1 Introduction

A cyclic listing of all proper colourings in which successive colourings differ at one vertex is a colouring Gray code, or equivalently a Hamilton cycle in the recolouring graph. Prescribing the first and last colourings asks for Hamilton-connectedness. Requiring a cycle of every length through each colouring asks for vertex-pancyclicity. We study these stronger spanning and cyclic properties for sparse graphs, with particular emphasis on sharp palette thresholds for planar classes.

All graphs are finite, simple and undirected, and may be disconnected. A graph is nonempty when it has at least one vertex. Put $[k] = \{1, \ldots, k\}$. The vertices of $\mathcal{R}_k(G)$ are the proper maps $\sigma : V(G) \to [k]$; two are adjacent when they differ on exactly one vertex of G. Colours are labelled, and a colouring need not use every colour. A cycle has at least three vertices. A graph is Hamilton-connected if every two distinct vertices are joined by a spanning path, and vertex-pancyclic if every vertex lies on a cycle of every length from three to the order of the graph.

For a class $\mathcal{C}$ of nonempty graphs, define

$$h(\mathcal{C}) = \min\{q : \mathcal{R}_k(G) \text{ is Hamiltonian for every } G \in \mathcal{C} \text{ and every } k \geq q\},$$
$$\mathrm{hc}(\mathcal{C}) = \min\{q : \mathcal{R}_k(G) \text{ is Hamilton-connected for every } G \in \mathcal{C} \text{ and every } k \geq q\},$$

when these sets are nonempty. These are uniform thresholds in the palette size; the proofs establish every larger palette separately. A graph is d-degenerate if each nonempty subgraph has a vertex of degree at most d. Write $\mathcal{D}_d$ for this class.

Choo and MacGillivray [3, Theorem 3.4] proved that $\mathcal{R}_k(G)$ is Hamiltonian for $k \geq d+3$ when G is d-degenerate. They also determined the uniform forest threshold and several individual tree and cycle cases [3, Sections 5–6]. Our first result strengthens their general Hamiltonian conclusion while keeping the same palette bound.

Theorem 1.1. *Let G be a nonempty d-degenerate graph. For every $k \geq d+3$, $\mathcal{R}_k(G)$ is Hamilton-connected and vertex-pancyclic. Moreover, $\mathrm{hc}(\mathcal{D}_d) = d+3$, and $d+3$ is also the least uniform palette bound for vertex-pancyclicity on $\mathcal{D}_d$.*

Write $\mathcal{P}$ for the planar graphs, $\mathcal{T}$ for the triangle-free planar graphs, $\mathcal{B}$ for the bipartite planar graphs, $\mathcal{P}_g$ for the planar graphs of girth at least g, and $\mathcal{F}$ for the forests. All these classes contain only nonempty graphs; forests have infinite girth.

Theorem 1.2. *The common thresholds $h = \mathrm{hc}$ are 7 for $\mathcal{P}$, 5 for $\mathcal{T}$ and $\mathcal{B}$, and 4 for $\mathcal{P}_6$ and $\mathcal{F}$.*

The large-palette cases follow from degeneracy. The critical palettes require endpoint-preserving restoration of separated low-degree vertices, and, in the triangle-free case, of an adjacent degree-three pair. The lower bounds use frozen colourings and the three-colour recolouring graph of a short path. The equality $h(\mathcal{F}) = 4$ already follows from [3]; here it accompanies the Hamilton-connected forest result.

At four colours the girth-five case requires different local configurations. Let $\mathcal{P}_5^{\leq 7}$ consist of the graphs in $\mathcal{P}_5$ with at most seven distinct 5-cycles. Let $\mathcal{S}_5$ consist of those in $\mathcal{P}_5$ whose every block has maximum degree at most three. Here blocks include bridges and isolated vertices as well as maximal two-connected subgraphs.

Theorem 1.3. *For every $G \in \mathcal{P}_5^{\leq 7} \cup \mathcal{S}_5$ and every $k \geq 4$, $\mathcal{R}_k(G)$ is Hamiltonian. Both classes have uniform Hamiltonian threshold four, and their Hamilton-connected thresholds lie between four and five.*

In particular, Theorem 1.3 includes all subcubic planar graphs of girth at least five, while allowing vertices of arbitrarily large total degree when that degree is distributed among blocks. Bartier et al. [1] proved that $\mathcal{R}_4(G)$ is connected for every $G \in \mathcal{P}_5$. The spanning-cycle and prescribed-endpoint questions remain stronger: for the full class we obtain

$$4 \leq h(\mathcal{P}_5) \leq 5, \qquad 4 \leq \mathrm{hc}(\mathcal{P}_5) \leq 5.$$

Table 1.1 records the results and these open cases.

Table 1.1: Uniform thresholds. An interval records an unresolved value.

Graph class	h	hc
Planar	7	7
Triangle-free planar	5	5
Bipartite planar	5	5
Planar, girth at least six	4	4
Forests	4	4
Planar, girth at least five, at most seven 5-cycles	4	$[4, 5]$
Planar, girth at least five, every block subcubic	4	$[4, 5]$
Planar, girth at least five	$[4, 5]$	$[4, 5]$

1.1 Proof strategy and organization

Deleting vertices partitions the larger recolouring graph into fibres above colourings of the smaller graph. At the degenerate bound these fibres are cliques of order at least three. We choose bridge colours between consecutive fibres to prescribe endpoints or cycle lengths. Section 2 proves Theorem 1.1, including the parity obstruction that makes its bound sharp. Section 3 develops the product and independent-set lifts used in the planar proofs.

Section 4 proves Theorem 1.2 by combining these lifts with planar degree counts. Section 5 proves Theorem 1.3. The few-cycle argument restores low-degree configurations. For the subcubic

and block-subcubic classes, we restore two separated pentagons. Their product fibres have at most four components. Local cycles retain protected edges at each interface; a bounded-degree spanning tree then joins all component cycles into one spanning cycle, for a bottom recolouring cycle of arbitrary length.

The supplement *Finite certificates for Hamiltonian thresholds in planar recolouring graphs* contains the literal paths, cycles, pools and interface records. Its Appendices A–D correspond to the adjacent-pair, path and pentagon steps. Section 7 specifies exactly what the accompanying verifiers check. The data certify local lemmas; the article supplies the arbitrary-order restoration arguments and inductions.

1.2 Notation

For a graph G, write $N_G(v)$ for the open neighbourhood, $N_G[v] = N_G(v) \cup \{v\}$, and $G - S$ for deletion of a vertex set S. The Cartesian product $X \square Y$ joins (x, y) to (x', y') when exactly one coordinate changes along an edge of its factor. A spanning path may be a singleton when its ends coincide in an intermediate construction. Local certificates use zero-based palettes $\{0, \dots, k-1\}$, obtained by relabelling $[k]$. Face lengths count edge-side incidences, so a bridge is counted twice.

2 Hamilton-connectedness and vertex-pancyclicity

We begin with the general result that supplies all large-palette statements.

Lemma 2.1 (Cyclic list selection)**.** *Let $m \geq 3$ and let $L_1, \dots, L_m$ be finite sets of size at least two. Unless m is odd and all L_i are the same two-element set, there are $x_i \in L_i$ with $x_i \neq x_{i+1}$ cyclically.*

Proof. If some L_i has at least three elements, choose successively along the path obtained by deleting that position, then choose the remaining element to avoid both end values. If all sets are binary and two consecutive sets differ, start with a value in their symmetric difference and continue greedily. If all are the same binary set, alternation is possible exactly when m is even. □

Lemma 2.2 (One-vertex lift)**.** *Let $H = G - v \neq \varnothing$ and suppose $k \geq d_G(v) + 3$. If $\mathcal{R}_k(H)$ has a Hamilton cycle, then $\mathcal{R}_k(G)$ has a Hamilton cycle.*

Proof. Write a Hamilton cycle of $\mathcal{R}_k(H)$ as $f_1, \dots, f_m, f_1$ and put $L_i = [k] \setminus f_i(N_G(v))$. Then $|L_i| \geq 3$ and $|L_i \cap L_{i+1}| \geq 2$. Lemma 2.1 gives cyclic choices $x_i \in L_i \cap L_{i+1}$ with consecutive values distinct. The extensions above f_i form a clique on L_i; traverse that clique by a Hamilton path from x_{i-1} to x_i, and use the edge with v fixed at x_i to cross to the next fibre. The exceptional all-identical binary case is impossible because every proper colouring of H occurs on the base cycle and a global colour permutation changes a neighbour of v to either binary colour. The fibre paths therefore concatenate to a spanning cycle. □

Lemma 2.3 (Bridge colours)**.** *Let $t \geq 1$, and let $S_0, \dots, S_{t-1}$ be finite sets of colours such that $|S_i| \geq 2$ for every i. Suppose that $|S_j| \geq 3$ for some $j \in \{0, \dots, t-1\}$. For any colours a and b, there exist colours $x_i \in S_i$ satisfying*

$$x_0 \neq a, \qquad x_i \neq x_{i-1} \quad (1 \leq i \leq t-1), \qquad x_{t-1} \neq b.$$

Proof. Set $x_{-1} = a$ and $x_t = b$. Choose $x_0, \dots, x_{j-1}$ forwards, avoiding the preceding colour, and $x_{t-1}, \dots, x_{j+1}$ backwards, avoiding the following colour. Each choice excludes only one colour. Finally choose $x_j \in S_j \setminus \{x_{j-1}, x_{j+1}\}$, which is possible since $|S_j| \geq 3$. The boundary assignments also cover $j = 0$, $j = t-1$, and $t = 1$. □

2.1 Clique fibres

For a vertex v of degree at most d with $H = G - v \neq \varnothing$, put

$$L(f) = [k] \setminus f(N_G(v)), \qquad F_f = \{f(c) : c \in L(f)\},$$

where $f(c)$ extends a proper colouring f of H by assigning colour c to v. The sets F_f partition $V(\mathcal{R}_k(G))$ into cliques. Whenever $k \geq d+3$, adjacent base colourings f, g satisfy

$$|L(f)| \geq 3, \qquad |L(f) \cap L(g)| \geq |L(f)| - 1 \geq 2. \tag{2.1}$$

The cross edges are exactly $f(c)g(c)$ for $c \in L(f) \cap L(g)$. If the base move changes a nonneighbour of v, then $L(f) = L(g)$. We use these fibres throughout the single-vertex lifts.

Lemma 2.4 (Single-vertex endpoint lift)**.** *Let $H = G - v \neq \varnothing$ and $k \geq d_G(v) + 3$. Suppose $Y = \mathcal{R}_k(H)$ is nonempty and Hamilton-connected, and v has a nonneighbour in H. If for every $f \in V(Y)$ and $c \in L(f)$ there is a neighbour g of f with $c \in L(g)$, then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. Fix a nonneighbour w of v. Colour permutations give all k colours at w among the states of Y; deleting at most one state leaves at least $k - 1 \geq 2$ colours. Thus every base path missing at most one state contains a move at w. Its consecutive intersections $S_i = L(f_i) \cap L(f_{i+1})$ have size at least two, and the move at w supplies one of size at least three. Lemma 2.3 now gives bridge colours avoiding any prescribed first and last colours, with distinct entrance and exit colours in every clique. Hamilton paths inside the cliques concatenate to a spanning path through those fibres with the prescribed endpoints.

Now prescribe distinct endpoints $\alpha(a)$ and $\beta(b)$ in $\mathcal{R}_k(G)$. If $\alpha \neq \beta$, lift a Hamilton path in Y between their restrictions.

If $\alpha = \beta$, choose $c \in L(\alpha) \setminus \{a, b\}$ and a neighbour g preserving c. Since $|V(Y)| \geq k \geq 3$, closing a Hamilton path between the ends of αg gives a Hamilton cycle through that edge. Let h be its other neighbour of α and choose $q \in (L(\alpha) \cap L(h)) \setminus \{c\}$. If $q = a$, exchange a, b and reverse the final path; thus assume $q \neq a$.

Partition F_α into paths $P : \alpha(a) \rightsquigarrow \alpha(c)$ and $P' : \alpha(q) \rightsquigarrow \alpha(b)$. Use a singleton for P' if $q = b$, and its endpoint edge otherwise; put all remaining vertices in P. Deleting α from the chosen base cycle leaves a path missing only that state. Lift it from $g(c)$ to $h(q)$ through all other fibres. The cross edges at c, q join this lifted path to P, P', giving the prescribed spanning path. □

Theorem 2.5 (Degenerate Hamilton-connectedness)**.** *Let G be a nonempty d-degenerate graph and $k \geq d + 3$. Then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. Induct on $n = |V(G)|$, starting with $\mathcal{R}_k(K_1) = K_k$. If $G = K_n$, then $\mathcal{R}_k(G)$ is the arrangement graph $A_{k,n}$. Since $k \geq n + 2$, it is Hamilton-connected by [5, Theorem 1].

Otherwise a minimum-degree vertex v has degree at most d and a nonneighbour. Put $H = G - v$; induction makes $\mathcal{R}_k(H)$ Hamilton-connected. For each base colouring f and $c \in L(f)$, choose a vertex of degree at most d in H. It has at least three available colours, so can be recoloured avoiding its present colour and c. This base move preserves c at v. Lemma 2.4 applies. □

Theorem 2.6 (Degenerate vertex-pancyclicity)**.** *Let $d \geq 0$ be an integer, let G be a nonempty d-degenerate graph, and let $k \geq d + 3$ be an integer. Then the recolouring graph $\mathcal{R}_k(G)$ is vertex-pancyclic.*

Proof. Induct on $|V(G)|$, starting with $\mathcal{R}_k(K_1) = K_k$. For larger G, choose v with $d_G(v) \le d$ and put $H = G - v$. Use the clique fibres above, and write $Y = \mathcal{R}_k(H)$, $M = |V(Y)|$, and $N = |V(\mathcal{R}_k(G))|$. Colour permutations give $M \ge k \ge 3$. Fix a state $x = \alpha(a)$ of $\mathcal{R}_k(G)$.

Closed-lifting claim. A base cycle $f_0, f_1, \dots, f_{s-1}, f_0$ through $f_0 = \alpha$ lifts to a cycle through x of every length in

$$[2s + 1,\ \textstyle\sum_i |F_{f_i}|]. \tag{2.2}$$

Put $S_i = L(f_i) \cap L(f_{i+1})$, with cyclic indices. Lemma 2.1 selects bridge colours $y_i \in S_i$ with $y_{i-1} \ne y_i$, unless s is odd and all S_i equal one binary set $\{p, q\}$.

We exclude this exception. By (2.1), each $L(f_i)$ then has size three, and

$$k - 3 = |f_i(N_G(v))| \le d_G(v) \le d \le k - 3.$$

Thus $d_G(v) = d = k - 3$, its neighbours have distinct colours, and none uses p or q. Every base move changes a neighbour of v; a nonneighbour move would leave an intersection of size three. Order the neighbours $u_1, \dots, u_d$. In the $(d+1)$-element palette $[k] \setminus \{p, q\}$, append the unique unused colour r_i to the tuple $(f_i(u_1), \dots, f_i(u_d))$. A base move exchanges one neighbour colour with r_i, reversing permutation parity. An odd number of such exchanges cannot return to the initial tuple. Hence the bridge colours exist.

Choose an m_0-vertex path in F_{f_0} between its bridge states and containing x, with $3 \le m_0 \le |F_{f_0}|$. In every other clique choose a bridge-to-bridge path on $2 \le m_i \le |F_{f_i}|$ vertices. The sums of these integer intervals are exactly (2.2). The cross edges with v fixed at y_i close these paths into the desired cycle.

The clique F_α already supplies a triangle through x. For lengths four, five and six, recolour a vertex of degree at most d in H to a colour avoiding its present colour, its neighbours, and a. At most $d + 2$ colours are excluded, so this gives a neighbour β of α with $a \in L(\beta)$. Choose $p \in (L(\alpha) \cap L(\beta)) \setminus \{a\}$, $q \in L(\alpha) \setminus \{a, p\}$, and $r \in L(\beta) \setminus \{a, p\}$. The following cyclic sequences have the required lengths:

$$\begin{aligned} 4:&\ \ \alpha(a), \alpha(p), \beta(p), \beta(a), \\ 5:&\ \ \alpha(a), \alpha(q), \alpha(p), \beta(p), \beta(a), \\ 6:&\ \ \alpha(a), \alpha(q), \alpha(p), \beta(p), \beta(r), \beta(a). \end{aligned}$$

For each $3 \le s \le M$, induction supplies a base s-cycle C_s through α. Put $T_s = \sum_{f \in V(C_s)} |F_f|$. The closed-lifting claim gives every length in $[2s + 1, T_s]$. These intervals overlap successively, since $T_s \ge 3s \ge 2s + 3$ for $s \ge 3$. Also $T_M = N$. They therefore cover every length from seven to N, completing the proof. □

Remark 2.7. Vertex-pancyclicity here cannot be strengthened to edge-pancyclicity (cycles of every length through each edge). For any $k \ge 4$, colour the leaves of the 1-degenerate star $G = K_{1,k-2}$ distinctly with $3, \dots, k$. The extensions with centre colours 1 and 2 form an edge of $\mathcal{R}_k(G)$ in no triangle: such a triangle would recolour only the centre, which has no third available colour.

Proposition 2.8 (Optimal uniform palette bound)**.** *For every $d \ge 0$, $\mathrm{hc}(\mathcal{D}_d) = d + 3$. The same uniform palette bound for vertex-pancyclicity cannot be reduced.*

Proof. The upper bounds are the preceding theorems. For $d \ge 1$, take $G = K_{d+1}$ and $k = d + 2$. Append the unique unused colour to each ordered colouring of G. The resulting permutation of $d + 2$ colours changes by a transposition at every recolouring. Thus $\mathcal{R}_{d+2}(G)$ is bipartite by permutation parity, with equal parts and even order $(d + 2)! \ge 6$. A spanning path must have endpoints in opposite parts, so the graph is not Hamilton-connected. It has no triangle, so it is not vertex-pancyclic. For $d = 0$, the two-isolated-vertex graph has $\mathcal{R}_2(G) = C_4$, which fails both properties. □

3 Endpoint lifting and restoration

A colouring of a deleted graph determines the available colours at the vertices to be restored. The resulting extension graph is a fibre. In all the constructions below, assignments retain their actual coordinate values: equal assignments in two different fibres are distinct vertices, and a cross edge joins their two copies. In particular, a move of one vertex of the deleted graph may change several available lists at once.

Lemma 3.1 (Products of complete graphs)**.** *Let $s \geq 1$ and $F = \square_{j=1}^{s} K_{A_j}$, where every A_j is finite and $|A_j| \geq 2$. If all lists have size two, F has a Hamilton path between two prescribed vertices if and only if those vertices lie in opposite bipartition classes. If at least one list has size at least three, F is Hamilton-connected.*

Proof. In the binary case F is the hypercube Q_s, whose bipartition classes have equal size. The endpoint condition is therefore necessary. For sufficiency, induct on s, starting with the single edge Q_1. Choose a coordinate on which the prescribed endpoints differ and split along it. Their projections into Q_{s-1} have the same parity. Choose a projected vertex of the opposite parity; use the two induction paths to and from its copies, joined by the edge between those copies.

For the second assertion, we first show that $Y \square K_r$ is Hamilton-connected whenever Y is Hamilton-connected and $|V(Y)| \geq 3$. Induct on r, starting with $r = 1$. If the two endpoints lie in different copies of Y, order the copies starting and ending at those copies. Choose successive bridge vertices in $V(Y)$, with the first different from the first endpoint projection, consecutive bridges different, and the last different from the last endpoint projection. Such choices exist because $|V(Y)| \geq 3$: choose greedily, leaving the last bridge until the end, when it has at most two forbidden values. Prescribed-end Hamilton paths inside the copies then join into a spanning path. If both endpoints lie in one copy, take a Hamilton path between them there and choose an edge xy of it. By induction the other $r - 1$ copies have a spanning path from a copy of x to a copy of y in the same one of those copies. Insert that path in place of xy, using the two cross edges. Start with a complete-graph factor of order at least three and add the remaining factors to obtain the claim. □

Lemma 3.2 (Endpoint product-fibre lift)**.** *Let $m \geq 3$, $s \geq 1$, and let*

$$F_i = \square_{j=1}^{s} K_{A_{i,j}}, \qquad |A_{i,j}| \geq 2, \qquad 0 \leq i < m,$$

be disjoint fibres arranged along a path. Join equal assignments in consecutive fibres. At every transition assume that at least one list is unchanged and that, for every coordinate,

$$|A_{i,j} \setminus A_{i+1,j}| \leq 1, \qquad |A_{i+1,j} \setminus A_{i,j}| \leq 1.$$

If an internal fibre F_r, $1 \leq r \leq m-2$, has a factor of order at least three, then every prescribed $p \in F_0$ and $q \in F_{m-1}$ are joined by a path covering all vertices of all fibres exactly once.

Proof. Write

$$J_i = \square_{j=1}^{s} K_{A_{i,j} \cap A_{i+1,j}}$$

for the graph of common assignments. The intersections are nonempty, and an unchanged coordinate supplies an edge of J_i.

Starting at p, record the reachable exit assignments at each interface. If a fibre has a reachable entrance a, an interface edge has an admissible Hamilton-path endpoint: in a binary fibre choose its endpoint of parity opposite to a, and in a nonbinary fibre choose an endpoint different from a. Lemma 3.1 therefore shows that the reachable set never becomes empty.

At F_r, its outgoing interface J_r still contains an edge after any one assignment is removed. Indeed, if its large coordinate is unchanged, J_r contains a triangle. Otherwise that coordinate has intersection size at least two, and a different unchanged coordinate supplies a $K_2\square K_2$. Take a reachable entrance a in F_r. Hamilton-connectedness of F_r makes every assignment in $V(J_r)\setminus\{a\}$ reachable; this set contains an entire edge.

An incoming reachable edge permits every outgoing assignment to be reached in the next fibre. In a binary fibre the two endpoints of the edge have opposite parities, so one has the parity needed for any specified outgoing assignment. In a nonbinary fibre they are distinct, so one differs from that assignment. Thus the reachable exits contain an edge at every subsequent interface. At the last fibre, one endpoint of the incoming edge can be joined to the specified q. Backtracking the finite choices supplies one Hamilton path in each fibre; the actual cross edges concatenate them into the required spanning path. □

Lemma 3.3 (Hamilton-connected independent-set restoration)**.** *Let S be a nonempty independent set, $H = G - S \neq \varnothing$, and $k \geq 4$. Suppose*

$$d_G(u) \leq k-2 \quad (u \in S), \qquad |N_G(x) \cap S| \leq |S| - 1 \quad (x \in V(H)).$$

Assume that $\mathcal{R}_k(H)$ is Hamilton-connected and that some proper k-colouring of H leaves at least three colours available at a vertex of S. If α, β are proper k-colourings of G with $\alpha|_H \neq \beta|_H$, then $\mathcal{R}_k(G)$ has an α–β Hamilton path.

Proof. Take a Hamilton path of $\mathcal{R}_k(H)$ between the two restriction colourings. For a base colouring f, the fibre is exactly

$$\square_{u\in S} K_{A_f(u)}, \qquad A_f(u) = [k] \setminus f(N_G(u)).$$

Every list is at least binary. A move at an actual vertex x of H leaves the list of at least one non-neighbour in S unchanged; every other list loses and gains at most one colour. These are the transition conditions of Lemma 3.2, even when several lists change together.

A colouring with a nonbinary fibre has at least k distinct colour permutations: fix a vertex of the nonempty H and send its colour to each of the k labels. Every resulting fibre is nonbinary. The base Hamilton path has only two endpoints, so one of these colourings is internal. Lemma 3.2 now lifts the path with the specified, actual colours of α and β on S. □

For use of this lemma, first choose a vertex w on which two prescribed colourings differ, and then choose S with $w \notin S$. This ensures different restriction colourings; the lemma does not claim to handle arbitrary endpoints in the same fibre.

Lemma 3.4 (Cyclic product-fibre lift)**.** *Let $m \geq 3$ disjoint fibres $F_i = \square_{j=1}^{s} K_{A_{i,j}}$, with $s \geq 1$ and $|A_{i,j}| \geq 2$, be cyclically ordered. Join equal assignments in consecutive fibres and define $J_i = \square_j K_{A_{i,j}\cap A_{i+1,j}}$, with indices modulo m. Suppose every J_i contains an edge, F_0 has a factor of order at least three, and there are distinct edges $e_0 \in E(J_0)$ and $e_{m-1} \in E(J_{m-1})$ (compared as actual assignment pairs). Then the layered graph has a Hamilton cycle.*

Proof. Choose an edge e_i in every remaining interface, and choose $z_0 \in V(e_0) \setminus V(e_{m-1})$. For $1 \leq i < m$, there is a bijection $\phi_i : V(e_{i-1}) \to V(e_i)$ such that each $z, \phi_i(z)$ is an admissible pair of Hamilton-path endpoints in F_i. In a binary fibre match opposite parities. Otherwise use a bijection with no fixed assignment, which always exists between two two-element sets: interchange equal sets, send a common member to the other target member when there is exactly one common member, and use either bijection when the sets are disjoint.

Put $z_i = \phi_i(z_{i-1})$. Traverse F_i from z_{i-1} to z_i for $1 \le i < m$. Since $z_{m-1} \in V(e_{m-1})$, $z_{m-1} \neq z_0$, so the nonbinary F_0 has a Hamilton path from z_{m-1} to z_0. Join the fibre paths by the cross edges at the assignments z_i. They cover every fibre once and form one cycle. □

Lemma 3.5 (Independent-set restoration)**.** *Let S be a nonempty independent set, $H = G - S \neq \varnothing$, and $k \ge 4$. Suppose*

$$d_G(u) \le k-2 \quad (u \in S), \qquad |N_G(x) \cap S| \le |S| - 1 \quad (x \in V(H)).$$

If $\mathcal{R}_k(H)$ is Hamiltonian and some proper k-colouring of H leaves at least three colours available at a vertex of S, then $\mathcal{R}_k(G)$ is Hamiltonian.

Proof. Take a Hamilton cycle of the base and use the same product fibres as in Lemma 3.3. Every interface contains an edge. Index a nonbinary fibre as F_0 and choose a large coordinate in it. At either adjacent interface, that coordinate supplies a triangle if unchanged, or a $K_2 \square K_2$ together with a different unchanged coordinate if it changes. Thus both adjacent interfaces have at least two edges, and distinct edges e_0, e_{m-1} can be chosen. Apply Lemma 3.4. □

Lemma 3.6 (State-degree criterion)**.** *Let $H = G - v \neq \varnothing$ and $k \ge \max\{4, d_G(v) + 3\}$. If v has a nonneighbour, $\mathcal{R}_k(H)$ is Hamilton-connected, and $\delta(\mathcal{R}_k(H)) > d_G(v)$, then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. For each base colouring f and $c \in L(f)$, at most $d_G(v)$ moves make c unavailable at v: only a neighbour of v recoloured to c can do so. The state-degree bound therefore supplies a neighbour preserving c. Apply Lemma 2.4. □

A Hamilton-connected graph of order at least four has minimum degree at least three. A vertex of degree two would, with its neighbours chosen as the two endpoints, force a spanning path to have only three vertices; degrees zero and one are excluded as well. Every nonempty colourable H has at least k states in $\mathcal{R}_k(H)$, by colour permutation. These observations will supply the state-degree condition in the low-degree inductions.

4 Hamilton-connected planar classes

4.1 All planar graphs

We first establish the marked structural statement needed to retain a coordinate on which the prescribed endpoint colourings differ.

Lemma 4.1 (Separated degree-five vertices with a marked vertex)**.** *If G is planar and $\delta(G) \ge 5$, then, for any $w \in V(G)$, there are two degree-five vertices different from w whose distance is at least three. Vertices in different components have infinite distance.*

Proof. Extend G, without adding vertices, to a simple plane triangulation T. A degree-five vertex in T still has degree five in G, and a pair at distance at least three in T has that property in G. Moreover $\delta(T) \ge 5$. It suffices to prove the assertion in T. Euler's formula gives at least twelve degree-five vertices.

For a degree-five vertex x, put $W = N_T[x]$, $U = V(T) \setminus W$, $r = |U|$, and let B be the vertices at distance exactly two from x; write $b = |B|$. The five neighbours of x form a cycle in their cyclic order, so $e_T(W) \ge 10$. Contract the connected W and simplify parallel edges. The resulting

plane graph has $r+1$ vertices and $e_T(U)+b$ edges. As $|V(T)| \geq 12$, the planar edge bound gives $e_T(U)+b \leq 3r-3$. Since $e(T) = 3r+12$,

$$\sum_{z \in U} d_T(z) = e_T(U) + e(T) - e_T(W) \leq 6r - b - 1.$$

Let F_x be the degree-five vertices at distance at least three from x. The same sum is exactly

$$6r - b - |F_x| + \sum_{z \in B} (d_T(z) - 5) + \sum_{\substack{z \in U \setminus B \\ d_T(z) \geq 6}} (d_T(z) - 6).$$

It follows that $|F_x| \geq 1$, and if $|F_x| = 1$, every vertex of B has degree five.

Suppose every pair of degree-five vertices at distance at least three contains w, and let L be all degree-five vertices except w. For every $x \in L$, the nonempty F_x must be $\{w\}$. Hence w is at distance at least three from L, and every vertex at distance exactly two from x belongs to L. Take a shortest path from w to L, ending at x. The vertex two steps before x is also in L and is closer to w, a contradiction. □

Theorem 4.2 (Planar Hamilton-connectedness)**.** *For every nonempty planar graph G and every $k \geq 7$, $\mathcal{R}_k(G)$ is Hamilton-connected. Consequently* $h(\text{Planar}) = \text{hc}(\text{Planar}) = 7$.

Proof. Prove the seven-colour assertion by induction on $|V(G)|$, allowing disconnected graphs throughout. If $|V(G)| \leq 5$, the graph is four-degenerate and Theorem 2.5 applies. For $|V(G)| \geq 6$, a vertex v of degree at most four has a non-neighbour. The smaller $H = G - v$ is Hamilton-connected by induction. For every seven-colouring f of H, which has at least five vertices,

$$d_{\mathcal{R}_7(H)}(f) = \sum_{x \in V(H)} (6 - |f(N_H(x))|) \geq 6|V(H)| - 2|E(H)| \geq 12 > d_G(v).$$

Lemma 3.6 restores v.

Otherwise $\delta(G) \geq 5$. For prescribed distinct colourings α, β, choose an actual vertex w with $\alpha(w) \neq \beta(w)$. Lemma 4.1 supplies $S = \{u, v\}$, avoiding w, with $d(u) = d(v) = 5$ and $d_G(u, v) \geq 3$. Thus S is independent, no outside vertex is adjacent to both members, and $H = G - S$ is smaller and nonempty. Choose two nonadjacent neighbours x, y of u; they exist since its five neighbours cannot induce a planar K_5. Properly six-colour the planar $H - \{x, y\}$, and give both x, y colour seven. This is a proper colouring of H in which the neighbours of u use at most four colours, leaving at least three available. The restriction colourings differ at w, so Lemma 3.3 gives an α–β Hamilton path.

For $k \geq 8$, planar five-degeneracy and Theorem 2.5 suffice. For sharpness, colour the six antipodal vertex pairs of the icosahedron with six different colours [2]. Each vertex sees all other five colours, so this is a frozen proper six-colouring; its recolouring graph has more than one state and is disconnected. Thus both thresholds are seven. □

Planar five-degeneracy and Theorem 2.6 also give vertex-pancyclicity for $k \geq 8$.

4.2 Triangle-free and bipartite planar graphs

The critical five-colour step needs an adjacent-pair fibre, which is not a product of complete graphs. We give its finite input separately from the arbitrary-size lifting argument.

For $A, B \subseteq \{0, 1, 2, 3, 4\}$, put

$$F(A, B) = \{(a, b) \in A \times B : a \neq b\},$$

with adjacency given by changing one coordinate. When $|A|, |B|$ are three or four, exchange the coordinates if necessary so that $|A| \le |B|$. There are exactly seven types $(|A|, |B|, |A \cap B|)$.

Lemma 4.3 (Adjacent-pair edge cycles and exits)**.** *For each of the seven types*

$$331, 332, 333, 342, 343, 443, 444,$$

every edge of $F(A, B)$ lies on a Hamilton cycle. The type 333 is itself a 6-cycle. The types $342, 343, 443, 444$ are Hamilton-connected. If J is the common state set of one of the interfaces allowed in Lemma 4.4, then every $p \in F(A, B)$ has a neighbour $q \in J$, and hence a p–q Hamilton path in $F(A, B)$.

Proof. Supplementary Appendix A gives one short Hamilton cycle for each of the seven representatives and checks its orbit under the independent row and column permutations that agree on $A \cap B$ (with the stated side exchange for type 332). The resulting edge orbits are the full edge sets. The 53 endpoint words in supplementary Appendix A prove the four Hamilton-connected types. Their symmetry coverage and an exact subset dynamic program verify the complete endpoint classification.

For the interface exit, exchange coordinates if needed so that B is unchanged, and put $A_0 = A \cap A'$. All allowed transitions, including the unchanged case, give $|A_0| \ge 2$ and $|B| \ge 3$. For $p = (a, b)$ with $a \in A_0$, choose (a, d) with $d \in B \setminus \{a, b\}$; otherwise choose (c, b) with $c \in A_0 \setminus \{b\}$. Either choice is a neighbour in J. Deleting this edge from a Hamilton cycle gives the required p–q Hamilton path. □

Lemma 4.4 (Adjacent-pair interfaces and reset)**.** *Arrange full fibres $F(A_i, B_i)$ along a base path. Each list has size three or four in a fixed five-colour palette. At each step either both lists stay unchanged, or exactly one changes. A changed list is either replaced by another three-element list differing in one colour, or changes between a three-element list and a containing four-element list. Join the copies of equal assignments at consecutive layers. If an internal layer has $|A_r| = |B_r| = 4$, there is a spanning path between any prescribed first-layer and last-layer states.*

Proof. First consider one allowed interface between $F(A, B)$ and $F(A', B')$, and write J for their common assignment set. The preceding lemma supplies, in both directions, a Hamilton path from every entrance state to some state of J.

Use this property to propagate nonempty sets of reachable exits from the prescribed first state. The internal 44 layer is Hamilton-connected, and its outgoing interface has at least nine states. Any reachable entrance p can therefore reach all interface states except possibly p, yielding at least two distinct reachable exits. The next layer still has one four-element list and is one of $342, 343, 443, 444$, hence is Hamilton-connected. Two distinct reachable entrances let every state of that layer be an exit. If it is the last layer, this includes the prescribed endpoint. Otherwise its entire outgoing interface is reachable.

Thereafter this property propagates: for a specified exit q in the current layer, apply the one-interface property in reverse to find a Hamilton-path entrance in its fully reachable incoming interface. The last prescribed endpoint is therefore reachable. Backtracking gives one Hamilton path in every full fibre, joined by actual cross edges, with no repetition or omission. □

Theorem 4.5 (Triangle-free planar Hamilton-connectedness)**.** *If G is a nonempty triangle-free planar graph, then $\mathcal{R}_k(G)$ is Hamilton-connected for every $k \ge 5$. For the class of nonempty triangle-free planar graphs, $h = \mathrm{hc} = 5$.*

Proof. For $k \geq 6$, use three-degeneracy and Theorem 2.5. For $k = 5$, induct on the order of G, permitting disconnected graphs. Orders at most three are handled by two-degeneracy and the same theorem. At larger orders a vertex v of degree at most two has a non-neighbour. The induction hypothesis gives a Hamilton-connected base of at least five states, thus of minimum degree at least three; Lemma 3.6 applies.

Assume now $\delta(G) \geq 3$. Let L be its degree-three vertices and $s = |L|$. The triangle-free planar edge bound gives

$$s \geq 8 + \sum_{d_G(z) \geq 4} (d_G(z) - 4).$$

Given distinct α, β, choose w on which they differ. There are two exhaustive cases.

If $L \setminus \{w\}$ contains adjacent u, v, denote their other neighbours by u_1, u_2 and v_1, v_2. The four actual outside vertices are distinct: a common neighbour of u and v would form a triangle. In $H = G - \{u, v\}$ the restriction endpoints still differ at w, so induction gives a base Hamilton path between them. Its fibre over f is

$$F(A_f, B_f), \qquad A_f = [5] \setminus \{f(u_1), f(u_2)\}, \quad B_f = [5] \setminus \{f(v_1), f(v_2)\}.$$

The lists have size three or four. One actual base move affects at most one of them and gives precisely an interface allowed in Lemma 4.4. Edges among the four outside vertices restrict the base colourings, but do not alter this statement.

There is a base colouring with both lists four-element: use Grötzsch's theorem [4] to three-colour $H - \{u_1, u_2, v_1, v_2\}$, then give u_1, u_2 colour four and v_1, v_2 colour five. Each pair is independent by triangle-freeness, and edges across the pairs have distinct colours. At least five global colour permutations give distinct double-four-list base states. Since the Hamilton path has only two endpoints, one is internal. Apply Lemma 4.4 with the specified local endpoint states to complete the restoration.

Otherwise $S = L \setminus \{w\}$ is independent and has at least $s - 1 \geq 7$ vertices. For any outside vertex z of degree at least four, the displayed Euler budget implies $d_G(z) \leq s - 4 < |S|$. The only possible degree-three outside vertex is w, whose degree is also less than $|S|$. Thus no outside vertex sees all of S. Choose $u \in S$ and two of its neighbours x, y. They are nonadjacent. Three-colour $H - \{x, y\}$ by Grötzsch's theorem and give both x, y colour five. At most two colours then occur on the three neighbours of u, so u has at least three available colours. All conditions of Lemma 3.3 hold, and w is retained; it gives the desired endpoint path.

For sharpness, on the cube Q_3 view the colours as $\mathbb{Z}_2^2$ and set

$$f(x, y, z) = x(1, 0) + y(0, 1) + z(1, 1).$$

The three neighbours of every vertex have all three other colours, so the colouring is frozen and the complete four-colour recolouring graph is disconnected. The upper and lower bounds give both thresholds. □

Corollary 4.6. *The class of nonempty bipartite planar graphs has* $h = \mathrm{hc} = 5$. *For every graph in this class,* $\mathcal{R}_k(G)$ *is vertex-pancyclic for* $k \geq 6$.

Proof. Bipartite planar graphs are triangle-free, and the cube belongs to this class. The vertex-pancyclic assertion follows from three-degeneracy and Theorem 2.6. □

The same vertex-pancyclic bound $k \geq 6$ holds for all triangle-free planar graphs. This conclusion uses the degenerate theorem separately from Hamilton-connectedness.

4.3 Girth at least six and forests

Lemma 4.7 (Separated degree-two vertices with a marked vertex)**.** *If G is planar, $g(G) \geq 6$, and $\delta(G) \geq 2$, then for any $w \in V(G)$ there are two degree-two vertices different from w whose distance is at least three.*

Proof. Every component has a cycle, and the girth-six Euler bound gives at least six degree-two vertices per component. If there are at least two components, choose such a vertex different from w in each of two components. Suppose G is connected and let n_2 be its number of degree-two vertices. From $2|E(G)| \leq 3|V(G)| - 6$,

$$n_2 \geq 6 + \sum_{d_G(z) \geq 3} (d_G(z) - 3).$$

Let L consist of all degree-two vertices except w; then $|L| \geq n_2 - 1 \geq 5$. Suppose all pairs in L have distance at most two.

If $u, v \in L$ are adjacent, let a, b be their respective other neighbours. The absence of cycles of length at most five implies that the intersection of the two closed distance-two balls is exactly $\{u, v, a, b\}$, and that $d_G(a, b) = 3$. Thus at most three of these vertices can lie in L, a contradiction.

Otherwise L is independent. Its two-element neighbourhoods are pairwise intersecting; they are distinct because there is no four-cycle. At least four distinct pairwise intersecting two-element sets have a common element: if $\{a, b\}, \{a, c\}$ occur and a third set omits a, it must be $\{b, c\}$, after which no fourth distinct set is possible. Hence some vertex z is adjacent to all of L. Then $d_G(z) \geq n_2 - 1 \geq 5$, whereas the Euler inequality gives

$$n_2 \geq 6 + d_G(z) - 3 \geq n_2 + 2,$$

a contradiction. □

Theorem 4.8 (Girth at least six)**.** *For every finite integer $g \geq 6$, the class of nonempty planar graphs of girth at least g has $h = \mathrm{hc} = 4$. Every graph G in this class has Hamilton-connected $\mathcal{R}_k(G)$ for $k \geq 4$, and vertex-pancyclic $\mathcal{R}_k(G)$ for $k \geq 5$.*

Proof. For $k \geq 5$, use two-degeneracy and Theorems 2.5 and 2.6. At $k = 4$, induct on the order, starting with orders at most two, which are forests and covered by the one-degenerate theorem. If $|V(G)| \geq 3$ and $d_G(v) \leq 1$, then v has a non-neighbour. The smaller nonempty base is Hamilton-connected by induction and has at least four states, hence minimum degree at least three. Restore v using Lemma 3.6.

Otherwise apply Lemma 4.7 to a vertex w at which the prescribed endpoint colourings differ. It gives separated degree-two vertices u, v, both different from w. Set $S = \{u, v\}$ and $H = G - S$. The set S is independent, no outside vertex sees both its members, and H is nonempty. The neighbours x, y of u are nonadjacent. The two-degenerate $H - \{x, y\}$ can be three-coloured; giving x, y colour four leaves three colours available at u. Lemma 3.3 now restores a Hamilton path between the actual prescribed states.

For the lower bound, take a cycle of length at least g divisible by three and repeat the colours $0, 1, 2$ around it. This is a frozen proper three-colouring, so its complete three-colour recolouring graph is disconnected. Both class thresholds are therefore four. □

Theorem 4.9 (Forests)**.** *The class of nonempty forests has $h = \mathrm{hc} = 4$. Every such forest F has Hamilton-connected and vertex-pancyclic $\mathcal{R}_k(F)$ for $k \geq 4$.*

Proof. The upper bounds follow from one-degeneracy and Theorems 2.5 and 2.6. For the Hamiltonian lower bound, $\mathcal{R}_3(P_3)$ has twelve states. Its six states using three distinct colours have degree two, so every edge incident with them is forced in a spanning cycle. For each of the three middle-vertex colours b, these forced edges form a 4-cycle

$$(a,b,a),(c,b,a),(c,b,c),(a,b,c), \qquad \{a,b,c\}=\{0,1,2\}.$$

The three disjoint forced cycles prevent a Hamilton cycle. For the Hamilton-connected lower bound, $\mathcal{R}_3(K_2)=C_6$ is already not Hamilton-connected. □

5 Hamiltonian four-colour subclasses at girth five

The reductions below splice Hamilton cycles and establish Hamiltonicity at four colours. They do not prescribe arbitrary endpoints, so the Hamilton-connected thresholds of these subclasses remain in [4, 5]. Supplementary Appendices B–D supply their local certificates.

5.1 Graphs with few five-cycles

Lemma 5.1 (Subdivided-star lift)**.** *Let $S=\{u_1,\dots,u_s\}$ be independent, $s\ge 3$, and suppose $N(u_i)=\{w,x_i\}$, where $N(w)=S$, $w\notin S$, and the x_i are distinct and belong to $J=G-(S\cup\{w\})$. If $\mathcal{R}_4(J)$ is Hamiltonian and some proper colouring of J gives the same colour to two distinct x_i, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. Take a Hamilton cycle C of $\mathcal{R}_4(J)$. Choose three distinct edges of C. At each edge at most one x_i changes, so an unchanged x_j supplies a colour a_r at the rth chosen edge. Choose a tree on the four colours whose three edges contain a_1,a_2,a_3, respectively (a star works when there is one repeated label; if the labels are all distinct, use the star at the fourth colour). Make four copies of C, indexed by the colour of w, and splice the corresponding copies along the three selected edges according to that colour tree. The result is a Hamilton cycle B of $\mathcal{R}_4(G-S)$ with the following control: whenever w changes from a to b along B, some x_j has colour a or b.

Over a base colouring f the u_i-lists are

$$A_i(f)=[4]\setminus\{f(w),f(x_i)\}.$$

They have size at least two. At a move in J, one list changes and another coordinate is unchanged. At a move of w, all intersections are $[4]\setminus\{a,b,f(x_i)\}$, and the control property leaves a coordinate with a two-element intersection. Thus the product-fibre splice applies at every step. At the base colouring with $f(x_p)=f(x_q)$, give w that same colour; the two corresponding lists have size three, so both interfaces at this fibre contain two distinct splice edges. The resulting cycle traverses all extensions and proves the lemma. □

Lemma 5.2 (Five-face identity)**.** *Let C be a connected plane graph of girth at least five and minimum degree at least two. Let p be the number of 5-faces, $Q=\sum_{|F|\ge 7}(|F|-6)$, $s=|\{v:d(v)=2\}|$, and $D=\sum_v(d(v)-3)$ over vertices of degree at least three. Then*

$$p=12+Q+2D-2s.$$

Every 5-face is a simple 5-cycle, and distinct faces give distinct cycles unless $C=C_5$.

Proof. A facial walk has no immediate reversal and contains a cycle, hence has length at least five. Counting face lengths gives $2e = 6f - p + Q$, while the degree sum is $2e = 3n - s + D$; with $n - e + f = 2$, subtraction yields the identity. A repeated vertex on a facial walk of length five would give a shorter cycle. If one 5-cycle bounds two faces, all its vertices have degree two and connectedness gives $C = C_5$. □

Corollary 5.3 (Two-degeneracy from seven pentagons)**.** *A planar graph of girth at least five with at most seven distinct* 5*-cycles is two-degenerate.*

Proof. Otherwise a connected subgraph has minimum degree at least three. In its plane embedding the identity has $s = 0$, and therefore at least twelve 5-faces, which are distinct 5-cycles. This contradicts the hypothesis. □

Theorem 5.4 (At most seven distinct five-cycles)**.** *Let G be a nonempty planar graph with girth at least five and at most seven distinct* 5*-cycles. Then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$; therefore the Hamiltonian threshold of this class is four. No four-colour Hamilton-connected conclusion is claimed.*

Proof. For $k = 4$ use induction on $|V(G)|$, allowing disconnected graphs. The one-vertex graph gives K_4. A vertex of degree at most one is restored by Lemma 2.2. If two degree-two vertices are adjacent, use Lemma 5.5. Otherwise the degree-two vertices form an independent set S. In a component containing a cycle, Lemma 5.2 and the bound $p \leq 7$ give $|S| \geq 3$; a tree component has a leaf and was already reduced.

If no vertex is adjacent to all of S, every vertex outside S has at most $|S| - 1$ neighbours in S. Choose $u \in S$ with neighbours x, y. They are nonadjacent by the girth condition. Three-colour $G - S - \{x, y\}$ greedily using Corollary 5.3, and colour both x, y with a fourth colour. This gives a base colouring with three colours available at u. Lemma 3.5 restores S.

Suppose that a vertex w is adjacent to all of S. In the component under consideration write $d(w) = |S| + t$. The identity and $D \geq d(w) - 3$ give $p \geq 6 + Q + 2t$, so $p \leq 7$ forces $t = 0$ and $N(w) = S$. Write $N(u_i) = \{w, x_i\}$. The x_i are distinct, and two are nonadjacent since $|S| \geq 3$ and the graph has no triangle. Put $J = G - (S \cup \{w\})$. Greedy three-colouring of $J - \{x_p, x_q\}$ followed by assigning the fourth colour to x_p, x_q supplies the hypothesis of Lemma 5.1; induction on J completes this case. For $k \geq 5$, apply the degenerate Hamiltonian theorem to the corollary above (and use the one-vertex lift at each deletion); this is a separate argument and assumes no monotonicity in k. □

5.2 Local degree-two reductions

Lemma 5.5 (Adjacent degree-two restoration)**.** *Let u, v be adjacent degree-two vertices, with $N(u) = \{v, x\}$ and $N(v) = \{u, y\}$, where $x \neq y$. If $\mathcal{R}_4(G - \{u, v\})$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. For a base colouring f, the extension fibre is $F(A, B)$, where $A = [4] \setminus \{f(x)\}$ and $B = [4] \setminus \{f(y)\}$. For $A = B = \{a, b, c\}$ use the cycle (ab, ac, bc, ba, ca, cb); for $A = \{a, b, c\}$, $B = \{a, b, d\}$ use $(ab, ad, bd, ba, ca, cd, cb)$.

In fact, any two fibre Hamilton cycles have at least two common edges when at most one list changes. For an unchanged fibre, equal lists give C_6, while unequal lists give four distinct forced edges at the two degree-two states using the common colours. By exchanging coordinates suppose only A changes. If one value equals $B = \{a, b, c\}$, the common fibre is the path (ab, ac, bc, ba); its two end edges are forced in both cycles. Otherwise let a, b be the old and new excluded colours, c the

fixed excluded colour and d the fourth colour. The common fibre has five vertices ca, cb, cd, da, db and six edges. The old outside states ba, bd occur consecutively because bd has degree two with neighbours ba, cd. Thus the old cycle uses exactly four common-fibre edges; the new cycle does likewise with ab, ad, cd. The cycles share at least $4 + 4 - 6 = 2$ edges.

Since $x \neq y$, a base step changes at most one list. Cut an edge of a base Hamilton cycle and process its fibres in path order. At each splice the current fibre has lost at most one edge and the next is fresh, so a common edge remains. Cutting its two copies and adding the two actual base-step edges merges the circles. The fibres partition all states, giving a spanning cycle. □

Lemma 5.6 (Separated degree-two vertices)**.** *Let G be planar of girth at least five and let u, v be nonadjacent vertices of degree two with $N(u) \cap N(v) = \varnothing$. If $H = G - \{u, v\}$ has Hamiltonian $\mathcal{R}_4(H)$, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. Set $S = \{u, v\}$. Every vertex of H has at most one neighbour in S. Let x, y be the neighbours of u; they are distinct and nonadjacent. The planar triangle-free graph $H - \{x, y\}$ admits a proper three-colouring by Grötzsch's theorem [4] (the empty graph has its empty colouring). Use colours $0, 1, 2$ there and give both x, y colour 3. This is a proper colouring of H leaving three colours available at u. The hypotheses of Lemma 3.5 hold with $k = 4$ and $|S| = 2$, so that lemma supplies the required Hamilton cycle. □

5.3 Restoring a $(2, 3, 2)$ path

The following reduction uses only a Hamilton cycle of the deleted graph. It processes an entire constant-boundary run together with its two neighbouring nonconstant fibres, retaining the edges needed for the next splice. The three external neighbours remain distinct actual vertices.

Proposition 5.7. *If $F = \mathcal{R}_3(P_3)$, then $F \,\square\, P_\ell$ is Hamiltonian for every $\ell \geq 2$.*

Proof. Partition F into its three centre-colour squares Q_b. Order each square as (q_0, q_1, q_2, q_3), where q_0, q_2 are its monochromatic-endpoint vertices. On $Q_b \square P_2$ use

$$(0, q_0), (1, q_0), (1, q_1), (0, q_1), (0, q_2), (1, q_2), (1, q_3), (0, q_3).$$

This cube cycle contains all four edges between layers 0 and 1. For larger ℓ, append each subsequent square by an ordinary edge splice using an internal edge of the newest layer. The initial newest layer has two internal edges; each appended layer retains three. This leaves all four original interlayer edges intact.

The three centre-colour squares of F are connected by three edges between monochromatic-endpoint vertices, one for each pair of centre colours. Choose a tree of two of these edges. Each chosen edge xy produces a square with vertices $(0, x), (1, x), (1, y), (0, y)$. Delete its two interlayer edges and add its two copies of xy, joining the corresponding strip cycles. The two selected interlayer edges at the middle strip are distinct. Two such splices give a single spanning cycle, for arbitrary $\ell \geq 2$. □

Let x, y, z be a path of degrees $2, 3, 2$. Its external neighbours, in that order, are p, q, r. Assume they are distinct and outside the path. For a bottom colouring with boundary $t = (f(p), f(q), f(r))$, write

$$B_t = \{(u, v, w) : u \neq v,\ v \neq w,\ u \neq t_0,\ v \neq t_1,\ w \neq t_2\}, \qquad u, v, w \in \{0, 1, 2, 3\}.$$

A boundary is *bad* if all three entries agree, and *good* otherwise. Up to colour permutations and path reversal the good types are $001, 010, 012$. Their orders are $6 + 4 + 4 = 14$, $9 + 4 + 4 = 17$, and $6 + 6 + 4 = 16$, respectively. The bad type 000 has order 12.

Good fibres and safe edges. The cycles D_E, D_M, D_A for the normalized boundaries $001, 010, 012$ are specified by literal cycles and capacity checks in supplementary Appendix B; the coordinates are uvw. Allow all their colour images and path reversals. Legality, cyclic adjacency, distinctness, and the counts above verify that each is a Hamilton cycle of its fibre.

For a bad boundary aaa, partition B_{aaa} into squares Q_c indexed by $c \neq a$, with fixed middle coordinate c. A good neighbour of aaa is obtained by changing position j from a to $b \neq a$. If j is an end position, then for each $c \notin \{a, b\}$ there is exactly one edge common to the good fibre and Q_c. Let P_t consist of these two edges. For middle-change or all-distinct boundaries set $P_t = \varnothing$. A good-cycle edge outside P_t is called *safe*. Directly from the supplementary path certificate, every allowed cycle has the following properties:

1. For an end change, it contains the unique edge in each Q_c, $c \notin \{a, b\}$, and two opposite edges of Q_b.
2. For a middle change it has no vertex in Q_b and contains three edges of each of the other two squares.

For $t = 001$, the two special edges are 123–323 and 132–232; the two edges in Q_1 are 313–213 and 212–312. For $t = 010$, the retained three-edge paths are $121, 321, 323, 123$ and $132, 232, 231, 131$. These descriptions also prove the assertions for every symmetry image.

Lemma 5.8 (Safe common edges)**.** *For adjacent good boundaries t, s, every allowed cycle at t and every allowed cycle at s have at least two common edges outside $P_t \cup P_s$.*

Proof. Partition the common fibre graph into its fixed-middle layers H_v and the edge set H_* changing the middle coordinate. In any block, if m edges remain outside $P_t \cup P_s$ and the two cycles retain x, y of them, they share at least $\max(0, x + y - m)$ safe edges. These bounds add over disjoint blocks. The supplementary path-kernel certificate classifies all allowed transitions and gives sufficient block counts for seven families, each yielding at least two.

Let a, b, c, d be distinct and use the selected cycles defined there. For $aab \to bab$, the common middle-b prism has no special edges. The E cycle retains one edge in each row triangle and two vertical edges. The cycle M_0 contains both these triangle edges, and M_1 both these vertical edges, giving two safe common edges in either case.

For $aba \to cba$, H_a has nine edges. Both cycles omit *bab–bac* and *bab–bad*: the M layer isolates *bab*, while the A path starts bab, dab. They retain three and five edges in this seven-edge universe, giving one common edge. In H_c they retain three and two of four edges, giving another. Both special sets are empty. Colour permutations and reversal transport all arguments, proving the assertion for every allowed labelled pair. □

Bad blocks.

Lemma 5.9 (Bad-block lifting)**.** *Consider consecutive fibres with boundaries*

$$t, \underbrace{aaa, \ldots, aaa}_{\ell \text{ times}}, s, \qquad \ell \geq 1,$$

where t, s are good and each differs from aaa in one coordinate. When $\ell = 1$ suppose $t \neq s$. Choose any allowed good cycles at the two ends, and reserve at most one safe edge in each. The layered graph on these fibres has a Hamilton cycle retaining both reserved edges. Only consecutive-layer matching edges are used.

Proof. Call a common edge with Q_c *available* if it belongs to the chosen good cycle and is not its reserved edge. An end-change good cycle has available edges in all three squares: the two singleton edges are special, hence cannot have been reserved, and at least one of its two Q_b edges remains. A middle-change good cycle has at least two available edges in each of its two supported squares.

First suppose $\ell = 1$. Let S_L, S_R be the sets of square colours supported by available edges of the two chosen good cycles. Then $|S_L|, |S_R| \geq 2$ and $S_L \cup S_R = [4] \setminus \{a\}$: the only exception could be two middle changes to the same colour, which would give $t = s$.

Choose a common square with different available edges on the two sides. If either change is at the middle, any $c \in S_L \cap S_R$ works, since that side offers at least two edges. If both changes are at ends, choose $c \notin \{a, b_L, b_R\}$, where b_L, b_R are their changed colours. Its two singleton edges are different: different ends give different coordinate directions, while changes at the same end have $b_L \neq b_R$ and give opposite square edges.

Start with the two good cycles and three bad squares. Attach the chosen square to both good cycles using these different edges, and attach each other square to a supporting good cycle. These four incidences form a tree. Cuts at a good cycle lie in different middle-colour squares, and the two bridge-square cuts are distinct. Along each tree edge delete the two selected internal edges and add the two matching edges. Each splice joins disjoint cycles into one; all reserved edges survive.

Now let $\ell \geq 2$. Each good end supports at least two squares, so choose an available left edge in Q_c and an available right edge in Q_d with $c \neq d$. Build the three square-strip cycles as in Proposition 5.7. The cube formula there permits either perfect matching of a square to be used in the first layer, by rotating its cyclic order; the last layer of a two-layer strip uses the other matching. Consequently its first layer can retain any specified square edge. For a two-layer strip the same is true at its last layer; our two requirements are on different strips and are independent.

If $\ell \geq 3$, a last-layer requirement is met by appending each new square with a splice edge different from the required edge. There are initially two choices and subsequently three in the current last layer, so this is always possible. The new last square retains all edges except the splice edge. First-layer horizontal edges and all four original vertical edges survive these extensions.

Join the three strips using the two vertical-edge switches in Proposition 5.7; these do not affect either required horizontal edge. Finally join the left and right good cycles to their selected strip edges by two ordinary edge splices. The result is one Hamilton cycle. The reserved edges were never cut. This proves the claim for every length, rather than just a bounded set of runs. □

Restoration over a base cycle.

Lemma 5.10 ($(2,3,2)$-path lifting)**.** *Suppose the three external neighbours of a degree-$(2,3,2)$ path are distinct and outside it. If the reconfiguration graph of the deleted graph is Hamiltonian at four colours, then so is $\mathcal{R}_4(G)$.*

Proof. Fix an arbitrary bottom Hamilton cycle. There must be a good vertex on it. Otherwise every boundary is constant; consecutive constant triples can only be equal, as at most one coordinate changes. The external colour would therefore be fixed throughout the cycle. A global colour permutation of a proper colouring supplies another proper colouring with a different external colour, a contradiction to Hamilton coverage. Thus even the startup uses only Hamiltonicity.

Every good bottom vertex has at most one bad neighbour in the entire bottom reconfiguration graph: a two-colour boundary has exactly one possible coordinate change making it constant, and an all-distinct boundary has none. The full neighbouring colouring, if proper, is uniquely determined by that change. Hence no good vertex on the Hamilton cycle can lie between two bad vertices. In particular there is an edge joining two good vertices.

Cut such an edge. In the resulting bottom Hamilton path, group each maximal bad run together with its two neighbouring good vertices; make each remaining good vertex a singleton block. These blocks are disjoint and partition the path. A bad run has constant boundary, and its two neighbours are distinct. If the run has length one their boundaries are different: otherwise both neighbours are obtained from the intervening colouring by the same one-vertex recolouring and would be the same bottom vertex.

For each labelled good boundary fix one allowed cycle once and for all. Consecutive blocks meet at a good–good boundary transition. If the boundary changes, Lemma 5.8 provides two common safe edges. If it stays unchanged, the fixed cycles coincide and have at least $14 - 2$ safe edges. Select one connecting edge for each interblock interface in path order. If a singleton block already has an incoming selected edge, choose a different outgoing one, which is possible from the two candidates. In a nonsingleton block the two selected edges are in its distinct end fibres, so cannot conflict.

Use Lemma 5.9 to construct each nonsingleton block cycle preserving its selected end edges; singleton blocks already have their fixed cycles. Splice the resulting disjoint block cycles along the selected interfaces. Each step joins two cycles into one, and every later selected edge is still present. The one-block case needs no final splice. All transitions are legal: a local step changes one path vertex, and a matching cross-edge performs its actual bottom recolouring while holding the local colouring fixed. Finally the fibres are disjoint and exhaust all proper colourings. Thus the resulting single cycle is Hamiltonian, including its closing edge. □

5.4 Protected pentagon restoration

Theorem 5.11. *Let G be a nonempty finite simple planar graph of girth at least five and maximum degree at most three. Then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. Disconnected graphs are allowed. The uniform Hamiltonian threshold of this class is exactly four, and its Hamilton-connected threshold lies between four and five.*

We prove a local restoration lemma using only Hamiltonicity of the smaller recolouring graph, and then supply the planar structure. Supplementary Appendices C–D specify the cycles, seeds, pools and interface records for 2,304 labelled pentagon boundaries and 313 standard-source one-entry interfaces. We first prove the protected-edge invariant, then use it in an arbitrary-order restoration.

Local certificates. Put $Q = \{0, 1, 2, 3\}$. A boundary t has five entries in Q, with at most one entry $*$. Its local graph F_t consists of the proper C_5 colourings x satisfying $x_i \neq t_i$; $*$ excludes no colour. In the certificates $*$ is encoded by 4. The eleven ordinary types have 590 states in total. The constant type has two components, each a 15-cycle; all other ordinary types are connected. Eight have the printed Hamilton cycles, and the two remaining types are 00001 and 00112. With a full list, the eleven types, after rotation, reflection and colour permutation, are

$$0000*,\ 0001*,\ 0010*,\ 0011*,\ 0012*,\ 0101*,\ 0102*,\ 0110*,\ 0112*,\ 0120*,\ 0123*.$$

Their printed Hamilton cycles have respectively $54, 60, 78, 62, 71, 88, 85, 74, 74, 83, 82$ states, a total of 811. Thus there are 23 component types, including the two constant components.

Lemma 5.12 (Ordered path blocks)**.** *Every one of the 23 component types B has a partition into paths $P_1, \ldots, P_r$ of order two or three, with an actual B edge joining P_i to P_{i+1} for each $i < r$.*

Proof. Twenty-one types have the printed Hamilton cycles, and 00112 has the 51-state Hamilton path in supplementary Appendix C. Delete a cycle edge where necessary, and divide the resulting path into consecutive two-vertex blocks, with one three-vertex block if its order is odd. For 00001,

supplementary Appendix C supplies the ordered path blocks and their actual connectors. The verifier checks coverage and every one-coordinate difference. □

For a Hamilton component A with specified cycle C_A, designate an ordered pool of five edges for either constant component, and eight edges otherwise. Let K_A be the other cycle edges. The first two pool edges join copies of C_A in a two- or three-row path product; the remaining edges supply at least two splice reserves. For a hard component, specify cycles of $A\square P_2$ and $A\square P_3$, and a set D_A of internal edges occurring in *every* row of both seeds. For each of 00001 and 00112, supplementary Appendices C–D specify a two-row cycle and derive its three-row seed by the same three-path insertion rule. Both use $|D_A| = 13$ and a two-edge pool. Put $K_A = D_A \setminus$ pool; the two distinct pool edges supply the splice reserves.

Here is the exact normalization convention. Enumerate $s = 0, \ldots, 4$, then $\epsilon = +1, -1$, then the permutations $p \in S_4$ in lexicographic order. Take the first standard type $n_i = p(t_{s+\epsilon i})$, retaining $*$. Restoration puts the standard entry x_i at position $s + \epsilon i$ with colour $p^{-1}(x_i)$. In particular a standard boundary first normalizes by the identity.

Lemma 5.13 (Protected-edge identities)**.** *Let the source be a standard component A of F_t, and change one non-star boundary entry. Normalize the target by the preceding rule. Every target component B whose restored state set meets A has an edge in common with K_A after restoration of K_B. For an unchanged boundary use the same component and the same map.*

Proof. The ordinary and starred orbit representatives cover $4^5 + 5 \cdot 4^4 = 2304$ labelled boundaries. The displayed cyclic words give their specified component cycles and the two-row hard seeds: their entries are precisely the legal words, without repetition, and successive entries, including last and first, differ in one coordinate. Both three-row seeds follow from the supplement's three-path insertion rule, using two edges and the displayed 40-state or 47-state path. Every listed common edge occurs in every row of both corresponding seeds. Removing the listed pools therefore defines the asserted protected sets. For the 313 standard-source interfaces, supplementary Appendix D gives 47 individual common edges, twenty-one source-selector rows covering 236 cases, and six constant-source families covering 30 cases. Endpoint membership in the specified templates and outside their pools proves each individual record. The selector rows use fixed protected incident edges against the opposite component's full degree, with the prescribed first normalizer and pools. The constant-source families use two normalizing directions and four protected target edges. Inverse coordinate and colour maps give every labelled case. This is a complete finite case proof; no finite-length gluing experiment infers a general rule. □

Products and overlaps.

Lemma 5.14. *For any two of the 23 component types A, B, the graph $A\square B$ has a Hamilton cycle retaining a copy of every edge of K_A in every physical B row.*

Proof. Use the ordered path blocks of Lemma 5.12 for B. On each block construct its product cycle with A. If A has its specified cycle C_A, join two copies using the first pool edge; for a three-row block use the first and second pool edges at its two interfaces. These cuts are distinct in the middle row. If A is hard, use its specified two- or three-row seed. Every initial cycle retains K_A and at least two common reserve edges in every physical row.

Choose two distinct reserve edges e, f. Join consecutive block cycles by cutting their two copies of e or f, alternating along the block chain, and adding the two cross edges supplied by the indicated actual edge of B. Each block participates in at most two joins. If both use the same physical row,

they cut different edges; thus every required copy is still present. Processing the blocks in order merges their disjoint cycles into one, covers every row, and never cuts K_A. □

Lemma 5.15. *For a fixed star position, or no star, let two boundaries be equal or differ in one non-star position. Any overlapping component pair has at least ten common vertices.*

Proof. An unchanged constant component has 15 vertices. Between a constant and nonconstant boundary, one additional colour is excluded at one coordinate of that 15-cycle. A cyclic permutation of its three colours preserves each component: an order-three action on a two-element component set is trivial. Hence each colour appears five times at each coordinate and precisely ten states remain. In every other case, both components are their entire local state sets. The intersection lists have size at least three except possibly one list of size two. Start at that position, proceed around the cycle, and finally avoid the two already coloured neighbours. This gives at least $2 \cdot 2^3 \cdot 1 = 16$ colourings. With no changed position the bound only improves. Full lists cause no difficulty. □

Consequently, if each of two boundaries changes at most one entry, normalize the target *relative to the source maps*. The two reference product cycles have at least ten common internal edges: one left protected edge from Lemma 5.13, copied at the ten or more common right states from Lemma 5.15. The two coordinate permutations and colour permutations may be independent. Actual synchronous changes are included in this stronger local classification; physical exterior vertices are never identified merely because they have equal colours.

Restoration over a base cycle.

Lemma 5.16. *Suppose G has girth at least five and contains two vertex-disjoint 5-cycles with no edge between them. Each cycle has vertices of degree three except for at most one vertex of degree two. Let H be obtained by deleting both cycles. If $\mathcal{R}_4(H)$ is Hamiltonian, so is $\mathcal{R}_4(G)$. Neither planarity nor three-colourability of H is assumed here.*

Proof. Each degree-three cycle vertex has one exterior neighbour; a degree-two vertex gives a full list. An exterior vertex meets a given 5-cycle at most once, since two attachments would create a 3- or 4-cycle. Thus a bottom recolouring changes at most one list in each factor. The two factors may change simultaneously. All exterior neighbours remain in H, which is nonempty.

Along a bottom Hamilton cycle, every ordinary pentagon has a nonconstant boundary somewhere. Otherwise consecutive constant five-tuples, differing in at most one entry, must be equal. Its entire boundary would stay fixed along the bottom cycle. A global colour permutation changing that colour supplies another proper colouring of H which the Hamilton cycle must visit, a contradiction. These nonconstant layers need not coincide for the two factors.

Make a component graph whose vertices are all product components in all bottom layers, joining components in consecutive layers when they have a common local state. There are at most four components per layer and every component-graph degree is at most eight. This graph is connected. Indeed, on each constant run the two 15-cycle components can be labelled consistently by a binary bit: consecutive constant boundaries are equal. A nonconstant layer forgets, or resets, that bit. A starred factor resets it at every layer. Take four bit-labelled copies of the bottom cycle and allow a flip of each bit at its reset layers. This graph is connected because each bit has a reset somewhere. Contract the reset edges within each layer. The result is exactly the component graph: intersections between nonconstant components are nonempty, each constant component meets its nonconstant neighbour, and an unchanged constant component meets precisely itself.

Choose a spanning tree of the component graph. Give the root one product cycle from Lemma 5.14 and record its two restoration maps. For a child, apply the inverse parent maps to the target

boundaries, normalize, then compose restoration maps. The target component is fixed by any common state. This makes the source a standard type for the protected-edge table, so the two *reference* product cycles share at least ten internal edges. The parent's component has tree degree at most eight and has so far lost at most seven internal edges. The child is fresh. Choose an unused common edge xy, delete its two copies and add the cross-layer edges at x and y. Each cross edge is the actual bottom recolouring with the complete local colouring unchanged. The two cycles become two paths joined into one cycle. Induction over the tree preserves one cycle, disjoint coverage and legality, and ends with all extensions of all bottom colourings. The final cycle spans the complete recolouring graph. □

5.5 Two separated five-faces

Lemma 5.17. *Every connected plane graph of girth at least five with $2 \le d(v) \le 3$ and at most one degree-two vertex contains two vertex-disjoint 5-cycles with no edge between them.*

Proof. For any connected cyclic plane graph of girth at least five and maximum degree three, let p and q count vertices of degrees one and two. Euler's identity, $2e = 3n - 2p - q$, and the face-length sum give

$$f_5 = 12 - 4p - 2q + \sum_{|F|\ge 7} (|F| - 6). \tag{*}$$

Every facial walk contains a cycle and so has length at least five; bridges count twice. A length-five facial walk is a simple 5-cycle. For the graph in the lemma, $p = 0, q \le 1$, hence $f_5 \ge 12 - 2q$.

If there is a bridge ab, take the side C not containing the unique degree-two vertex, if it exists. In C, only a has degree two and all other vertices have degree three. This side is cyclic, and (*) gives at least ten 5-faces. At most two contain a. Choose a 5-cycle S avoiding a. The other bridge side has either at most two degree-two vertices, or one degree-one vertex when b was the original degree-two vertex. All other vertices have degree three. The degree sum rules out a tree and (*) gives at least eight 5-faces. Any such cycle P is disjoint from S and has no edge to it, because ab was the only edge between the sides.

Assume now there is no bridge. The subcubic graph is 2-connected: at a cutvertex every branch would require at least two incident edges to avoid a bridge, forcing degree at least four. Take a facial 5-cycle S and let B be its exterior neighbours. Girth makes them distinct; $b = |B|$ is five, or four when S contains the unique degree-two vertex z. Set $H = G - S$, embedded with the deleted region outside. Every member of B is on that outside face. Its degree drops from three to two, except that $z \in B$ would become a leaf. There is at most one leaf and no isolated vertex, so every component of H is cyclic.

In one such component, let r count its members of B, $q' \le 1$ its original degree-two vertices, and ℓ its outside facial length. Formula (*) after removing the outside-face contribution is

$$f_5^{\text{in}} = \ell + 6 - 2r - 2q' + L, \qquad L = \sum_{\substack{F \text{ internal} \\ |F|\ge 7}} (|F| - 6) \ge 0. \tag{**}$$

Even when $z \in B$, the degree deficit $4p + 2q$ remains $2r + 2q'$, so the formula still applies. Here $\ell \ge 5$. A member of B belongs to at most one internal 5-face: its degree is at most two, with an outside face angle, and a leaf belongs to none. If H is disconnected, some component has $r \le 2$ since $|B| \le 5$. By (**) it has at least $9 - 2r > r$ internal 5-faces. One avoids B, and is the required P.

Suppose H is connected. The b exterior edges partition the outside angles along S into b face portions, so at most b other faces are involved. Deleting five vertices and $5 + b$ edges decreases the

face count by b; thus these b neighbours $F_1, \dots, F_b$ are distinct. Their merger has outside length

$$\ell = \sum_{i=1}^{b} |F_i| - (2b + 5), \tag{***}$$

because the five cycle edges are removed once and the b outgoing edges twice from that sum.

If $q = 0$, any S has $b = 5$, $\ell \geq 10$, $q' = 0$. Formula (**) gives at least six internal 5-faces, more than $|B|$. If z lies on a 5-face, choose that face as S. Now $b = 4$, $\ell \geq 7$, $q' = 0$, and there are at least five, again more than $|B|$.

It remains that z lies on no 5-face. There is a 5-face adjacent to two distinct non-5-faces. To see this, consider the plane dual. It is loopless because the original graph is bridgeless. All its faces are triangles except one digon corresponding to z. The distinct endpoints of that digon both represent non-5-faces. Let W be the set of all such dual vertices. If every vertex outside W had at most one distinct neighbour in W, a triangular face incident with a W–W edge would have its third vertex in W. Starting with a digon edge and going around either endpoint therefore shows that all its neighbours are in W: a triangular face propagates the property to the next neighbour, and the digon merely repeats the same neighbour. Repeating at any reached neighbour makes W closed under adjacency. Dual connectivity would force all vertices into W, contradicting $f_5 \geq 10$. This proves the assertion. Choose the resulting 5-face S. If its deletion disconnects the graph, use the case already proved; otherwise $b = 5, q' = 1$ and two distinct neighbours in (***) have length at least six. Thus $\ell \geq 12$ and (**) again gives at least six internal 5-faces. In every case one avoids B, making it disjoint and nonadjacent to S. □

Completing the subcubic induction.

Proof of Theorem 5.11. Induct on the number of vertices, allowing disconnected graphs. The one-vertex case is $\mathcal{R}_4(G) = K_4$. A vertex of degree at most one is handled by Lemma 2.2. If there are two degree-two vertices, adjacent ones use Lemma 5.5. Ones at distance two use Lemma 5.10, unless their middle vertex has degree two, when an adjacent pair already applies. Ones at distance at least three use Lemma 5.6, whose spare-colour witness uses Grötzsch's theorem [4]; the new pentagon lift does not. Pairs in different components have disjoint neighbourhoods and are covered by the same separated-pair argument.

The remaining graph has minimum degree at least two and at most one degree-two vertex. Apply Lemma 5.17 inside any component. Delete its two separated pentagons. The remaining graph is nonempty and belongs to the same class, so induction gives its four-colour Hamilton cycle. Lemma 5.16 restores both pentagons. This proves the assertion at four colours. For every $k \geq 5$ use the triangle-free planar theorem proved earlier, with larger colours separately supplied there by degeneracy. The frozen 012012 colouring of C_6 gives the Hamiltonian threshold lower bound four. □

5.6 Graphs with subcubic blocks

Theorem 5.18 (Subcubic blocks)**.** *Let G be a nonempty finite simple planar graph of girth at least five. If every block B satisfies $\Delta(B) \leq 3$, then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. Cutvertices may have arbitrarily large degree in G. The uniform Hamiltonian threshold of this class is exactly four. Its Hamilton-connected threshold lies between four and five.*

Proof. Treat bridges as two-vertex blocks and isolated vertices as singleton blocks. The class is closed under vertex deletion: a nontrivial two-connected block of a subgraph is a two-connected

subgraph of G, hence is contained in one original block, with no larger internal degrees. Planarity and girth are also preserved.

Products of Hamiltonian graphs are Hamiltonian: copy one factor cycle along a spanning path of another factor cycle, and merge successive copies using two alternating, distinct internal edges. This also handles disconnected original graphs. Choose a smallest four-colour counterexample, which may therefore be assumed connected. The singleton base is K_4, and the single-vertex lift excludes vertices of degree at most one. If there is only one block, the previous subcubic girth-five theorem applies. Thus assume minimum degree at least two and at least one cutvertex.

Lemmas 5.5, 5.10, and 5.6 exclude adjacent degree-two vertices, a $(2, 3, 2)$ path, and two degree-two vertices at distance at least three. Consequently any two degree-two vertices not already giving a reduction must have a common neighbour of degree at least four in G.

The block-cut tree has at least two endblocks. No endblock is a bridge, since its noncut endpoint would be a leaf. Let B be an endblock and x its unique cutvertex of G. Every vertex of $B - x$ has its full G-degree in B, hence degree two or three; also $d_B(x) \in \{2, 3\}$. Let q count the degree-two vertices of $B - x$ and put $\varepsilon = [d_B(x) = 2]$. If $q \geq 2$, any pair of these vertices has a common neighbour of G-degree at least four by the excluded reductions. Their neighbours all lie in B, so only x can be this common neighbour. Every one of the q vertices is therefore adjacent to x, giving $q \leq d_B(x) = 3 - \varepsilon$. Together with the immediate $q \leq 1$ case, this proves $q + \varepsilon \leq 3$.

Euler's identity in the two-connected block B now gives

$$f_5(B) = 12 - 2(q + \varepsilon) + \sum_{|F| \geq 7} (|F| - 6) \geq 6.$$

Each five-face is a simple pentagon, while at most $d_B(x) \leq 3$ faces meet x, so choose one avoiding x. If $q \leq 1$, it contains at most one degree-two vertex of G. If $q \geq 2$, a cycle through any of the q vertices must use its edge to x, so the chosen pentagon contains none. All its other vertices have G-degree three.

Choose these pentagons in two distinct endblocks. Their vertices are noncutvertices of G and belong to their respective endblocks only, so the pentagons are disjoint and have no edge between them. The established protected-edge lift for two separated pentagons applies, including its allowance of one degree-two vertex per pentagon and simultaneous exterior list changes. The smaller graph remains in the class and has a bottom Hamilton cycle by induction. Restoration contradicts minimality.

For every $k \geq 5$ separately use the established triangle-free planar theorem. Finally C_6 belongs to the class and its periodic 012012 colouring is frozen in $\mathcal{R}_3(C_6)$, giving the lower bound four. □

In particular, a remaining minimum counterexample for general planar girth-five four-colour Hamiltonicity must contain a block with a vertex of degree at least four *inside that block*. High total degree accumulated only at cutvertices is insufficient. This does not assert that every minimum counterexample is two-connected.

6 The girth-five frontier

The threshold intervals in Table 1.1 leave two questions: does every $G \in \mathcal{P}_5$ have a Hamiltonian $\mathcal{R}_4(G)$, and is every such recolouring graph Hamilton-connected? The latter question also remains open for the four-colour subclasses of Theorem 1.3. Known connectivity [1] does not settle either question. Theorem 5.18 shows that a minimum Hamiltonian counterexample must contain a block with internal maximum degree at least four.

7 Finite certificates and reproducibility

The supplementary source `article_certificates.tex` is the input for all three exact verifiers. The words, pools and selectors are read from that source. Missing blocks, duplicate records, illegal transitions, incomplete state coverage and failed protection conditions cause the check to fail. No solver output or earlier computation report is an input to these checks.

Table 7.1: Finite checks supporting the local restoration lemmas.

Supplement	Checked objects and coverage
Appendix A	Seven edge-covering cycle orbits; 53 endpoint words with exact subset-DP cross-check; 225 labelled list pairs and 3,225 interfaces.
Appendix B	Three normalized path cycles; 60 good boundaries and all 504 directed adjacent good-boundary changes; safe edges for every allowed cycle pair and support at bad neighbours.
Appendices C–D	23 complete pentagon components; 2,304 labelled boundaries; ordered path blocks; both hard two- and three-row seeds; pool membership and protection in every row; all 313 interfaces, partitioned as $47 + 236 + 30$.

Run `python3 supplement/verify_all.py` from the article directory. The scripts require only the Python standard library and use exact integer and finite-set operations. The runner also checks rejection of corrupted cycles, missing certificate blocks, invalid interface edges and altered selector data. Its JSON reports record input hashes, coverage and the Python version; the distribution includes commands for rebuilding both PDFs and a file manifest.

The finite verification establishes the local certificate claims in Table 7.1. The path lifting, arbitrary bad-run construction, protected products, spanning-tree splicing and structural inductions are analytic arguments in the article. Thus the finite checks have no bound on the order of G as an input premise, and the general girth-five questions stated above remain open.

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